

## Review article

# Ketamine versus electroconvulsive therapy for major depressive episode: An updated systematic review and non-inferiority meta-analysis

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## ABSTRACT

We conducted a systematic review and meta-analysis to investigate the comparative effectiveness of ketamine versus electroconvulsive therapy (ECT) for the treatment of major depressive episodes (MDEs). PubMed, EMBASE and Cochrane Library databases were systematically searched for randomized controlled trials (RCTs) comparing ketamine and ECT for MDE. The primary outcome was response rate, for which we prespecified a non-inferiority margin of -0.1, based on the largest and most recent RCT. Response was defined as a reduction of at least 50 % in the depression scale score. Six RCTs met the inclusion criteria, comprising 655 patients. In the overall population, ketamine was not non-inferior to ECT in response rate (RD -0.10; 95 % CI -0.26 to 0.05;  $p = 0.198$ ;  $I^2 = 72$  %). The ECT group had a higher reduction in depression scores, but without difference in remission and relapse rates. Regarding safety outcomes, ketamine had better posttreatment cognition scores and reduced muscle pain rate compared with ECT, albeit with an increased rate of dissociative symptoms. In a subanalysis with only inpatients, ketamine was inferior to ECT in response rate (RD -0.15; 95 % CI -0.27 to -0.03;  $p = 0.014$ ;  $I^2 = 25$  %), remission, and change in depression scores. These findings support the use of ECT over ketamine for inpatients. Further RCTs are warranted to clarify the comparative effect of these treatments for outpatients.

## Abbreviations

Major Depressive Disorder (MDD) 3  
Major Depressive Episode (MDE) 5  
Major Depressive Episodes (MDEs) 1  
Electroconvulsive Therapy (ECT) 44  
Randomized Controlled Trials (RCTs) 11  
Individual Patient Data (IPD) 1  
Montgomery-Åsberg Depression Rating Scale (MADRS) 1  
Hamilton Depression Rating Scale (HDRS) 1  
Risk of Bias tool version 2 (RoB 2) 1  
Risk difference (RD) 3

Risk ratios (RRs) 9

Standardized mean difference (SMD) 3

Confidence intervals (CIs) 15

Battery for ECT Related Cognitive Deficits (B4ECTReCoDe) 1

## 1. Introduction

Major depressive disorder (MDD) is one of the most prevalent psychiatric disorders worldwide, and the leading cause of mental-health-related disability (Friedrich, 2017; Gutiérrez-Rojas et al., 2020). MDD is associated with a significant and growing economic burden in the USA, accounting for 326.2 billion dollars in direct and indirect costs in

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2018 (Greenberg et al., 2021).

A major depressive episode (MDE) is defined as a period of at least two weeks in which the individual experiences a depressed mood or a loss of interest or pleasure in previously enjoyable activities (American Psychiatric Association, 2013). Although antidepressants are mostly effective for the treatment of MDE, more than 30 % of patients treated with pharmacotherapy do not reach remission of depressive symptoms, even after several trials (Rush et al., 2006).

Electroconvulsive therapy (ECT) is the gold standard treatment for acute depressive episodes, especially in patients with treatment-resistant depression (Gelenberg et al., 2010). It is also indicated for patients at risk of suicide, who require a rapid response (Kellner et al., 2005). ECT is administered 2–3 times a week, with an average number of sessions ranging from 6 to 18, and its mechanism of action is centered on the modulation of the functional connectivity of default mode and somatomotor networks, reducing depressive symptoms (American Psychiatric Association, 2001; Charlson et al., 2012; Guo et al., 2024). However, the limited availability and cognitive impairment associated with ECT underscores the need for alternative treatments with similar efficacy and better safety and acceptability profiles (Espinoza and Kellner, 2022; Husain et al., 2004; Lisanby, 2007).

Ketamine, an N-methyl-D-aspartate receptor antagonist used as a general anesthetic and analgesic, has been considered an option for major depressive episodes (MDEs) due to its glutamate-modulating action (Gao et al., 2016; Husain et al., 2004; Rasmussen et al., 2013). Furthermore, ketamine is not associated with significant cognitive adverse effects, and has fewer implementation barriers than ECT, motivating several studies to compare these treatments (Allen et al., 2015; Anand et al., 2023; Basso et al., 2020; Ekstrand et al., 2022; Ghasemi et al., 2014; Kheirabadi et al., 2020; Loureiro et al., 2020; Sharma et al., 2020; Smith-Apelidoorn et al., 2022). However, the relative efficacy of ketamine versus ECT remains uncertain, and the Food and Drug Administration currently does not authorize its use for the treatment of depression (Nikayin et al., 2022).

Although previous meta-analyses have compared ketamine with ECT and found superiority of ECT over ketamine, the recent ELEKT-D trial showed that ketamine was non-inferior to ECT (Anand et al., 2023; Menon et al., 2023; Rhee et al., 2022). Given this uncertainty, we performed a systematic review and non-inferiority meta-analysis of randomized controlled trials (RCTs) comparing ketamine versus ECT for the treatment of MDE.

## 2. Methods

This systematic review and meta-analysis was performed and reported according to the Cochrane Collaboration Handbook for Systematic Reviews of Interventions and Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) reporting guidelines (Appendix 1–2 in the Supplement) (Liberati et al., 2009; *The Handbook of Research Synthesis and Meta-Analysis*, 2019). The prospective protocol of this meta-analysis was uploaded to the International Prospective Register of Systematic Reviews (CRD42023446785).

### 2.1. Search strategy and inclusion criteria

A systematic search was conducted by two independent authors (A.B. C.P. and I.A.R.) in PubMed, the Cochrane Library, and Embase for studies published from inception to June 2023. The search terms used included “ketamine”, “electroconvulsive”, and “depression”. A highly sensitive search strategy for RCTs was used in PubMed and Embase (McKibbin et al., 2009). The complete search strategy used in each database is provided in Supplemental Table 1. Moreover, we performed backward snowballing (i.e., review of references) of the included articles. We restricted inclusion to studies that met all the following criteria: (1) RCTs; (2) comparing ketamine with ECT; and (3) among patients with a diagnosis of MDE. The exclusion criteria were (1)

non-randomized studies; (2) nonhuman studies; and (3) absence of standardized measures for depression. There were no language or year publication restrictions. After the initial search, two authors independently removed the duplicates, screened titles and abstracts, and evaluated the articles in full for eligibility. Discrepancies were resolved by a consensus-based discussion.

### 2.2. Data extraction

Primarily, data were extracted from published articles and, if available, from individual patient data (IPD) conceded by the authors. Otherwise, we retrieved data from prior meta-analyses. The extracted data included population baseline characteristics (sample size, sex, presence of psychotic features, and admission status), specifications of each treatment (number of sessions, ketamine dose, and ECT electrode placement), and definitions and time frames of the outcomes.

### 2.3. Outcomes

Our prespecified primary outcome was response rate, defined as the proportion of patients achieving a reduction of  $\geq 50$  % in the depression scale score used in each study. To standardize the effect measures, we prioritized extracting data from the Montgomery-Åsberg Depression Rating Scale (MADRS). If not possible, the Hamilton Depression Rating Scale (HDRS) was used as an alternative. A non-inferiority margin of  $-0.1$  in the risk difference for the response rate was predefined, based on the largest and most recent study (Anand et al., 2023).

The secondary outcomes were change in depression scores, remission rate, and relapse rates at 1, 3, and 6 months. The safety outcomes included posttreatment cognition scores, muscle pain, headache, and dissociative symptoms. We also conducted subanalyses based on psychotic features and admission status. A comprehensive description of each study definition and scales used for primary and secondary outcomes is provided in Supplemental Table 2.

### 2.4. Risk of bias assessment

Two independent authors used the Cochrane Collaboration Risk of Bias tool version 2 (RoB 2) to assess the risk of bias in each individual RCT (Sterne et al., 2019). Comparison-adjusted funnel plots were used to assess small study effects for the primary outcomes (Sterne et al., 2011). However, Egger's regression test was not possible due to the small number of studies ( $n < 10$ ) (Higgins et al., 2019).

### 2.5. Statistical analysis

For the primary outcome, the treatment effect was pooled with risk difference (RD) due to the availability of non-inferiority margins given by individual studies. For other binary outcomes, treatment effects were pooled using risk ratios (RRs). For continuous outcomes, we used the Hedges g standardized mean difference (SMD), due to the differences between the scales used in the studies. To enhance the interpretability of this measure, we used the Cohen definition of small, moderate, and large effect sizes (Cohen, 2013; Higgins et al., 2019). DerSimonian and Laird random effects model was used for all the outcomes to calculate the overall effect sizes with 95 % confidence intervals (CIs). Analysis was performed using the Maentel-Haenszel estimates for the dichotomous outcomes and the inverse variance statistic method for the continuous outcomes. The heterogeneity between the individual study results was analyzed with the Cochrane Q test and  $I^2$  statistics. Additionally, we conducted a meta-regression analysis to evaluate the influence of the mean age across studies on the primary outcome. We used R version 4.3.1 software for the statistical analyses (Shim and Kim, 2019).

### 3. Results

As detailed in Fig. 1, our initial search yielded 960 records. After duplicates and nonrelated studies were removed, 75 articles remained and were assessed for full-text review according to the eligibility criteria. Of those, six RCTs were included, comprising 655 patients, of whom 348 (53.1 %) were randomized to receive ketamine and 307 (46.9 %) to undergo ECT. Among the included studies, 2 were non-inferiority trials (Anand et al., 2023; Ekstrand et al., 2022), and 4 had superiority designs (Ghasemi et al., 2014; Kheirabadi et al., 2020, 2019; Sharma et al., 2020). All studies were conducted exclusively in patients over 18 years old, diagnosed with MDD, who were candidates for ECT. Five of the studies included only hospitalized patients, (Ekstrand et al., 2022; Ghasemi et al., 2014; Kheirabadi et al., 2020, 2019; Sharma et al., 2020), while one had a primarily outpatient population. (Anand et al., 2023). Four trials were restricted to patients without psychotic features (Anand et al., 2023; Ghasemi et al., 2014; Kheirabadi et al., 2020, 2019).

The number of doses administered in the ketamine group ranged from 6 to 9 among the included studies. Five RCTs employed intravenous administration, in the dose of 0.5 mg/kg (Anand et al., 2023; Ekstrand et al., 2022; Ghasemi et al., 2014; Kheirabadi et al., 2019; Sharma et al., 2020). One study involved both oral (0.5 mg/kg) and intramuscular (1 mg/kg) administration routes (Kheirabadi et al., 2020). ECT was conducted with bilateral electrode placement in three trials (Ghasemi et al., 2014; Kheirabadi et al., 2020, 2019), unilateral electrode placement in one trial (Anand et al., 2023), and two trials used both (Ekstrand et al., 2022; Sharma et al., 2020). The number of sessions in the ECT group ranged from 1 to 12. The follow-up duration ranged from one to four weeks, and two studies conducted long-term follow-up with patients with successful results in the initial treatment phase

(Anand et al., 2023; Ekstrand et al., 2022). A comprehensive description of the baseline patient and study characteristics, diagnostic and inclusion criteria, and treatment specifications can be found in Table 1 and in Supplemental Tables 3, 4, and 5.

#### 3.1. Response rate

The response rate was not significantly different between the groups ( $RD$ ,  $-0.10$ ; 95 %  $CI$ ,  $-0.26$  to  $0.05$ ;  $p = 0.198$ ;  $I^2 = 72$  %) (Fig. 2A). Heterogeneity for this outcome was moderate. Among patients without psychotic features, ketamine and ECT had similar results ( $RD$ ,  $-0.03$ ; 95 %  $CI$ ,  $-0.22$  to  $0.16$ ;  $p = 0.749$ ;  $I^2 = 59$  %) (Supplemental Figure 1). In hospitalized patients, ketamine was inferior to ECT ( $RD$ ,  $-0.15$ ; 95 %  $CI$ ,  $-0.27$  to  $-0.03$ ;  $p = 0.014$ ;  $I^2 = 25$  %) (Fig. 2B), with a low heterogeneity. A meta-regression analysis suggested no significant interaction between response rate and mean patient age ( $coefficient = 0.005$ ; 95 %  $CI$ ,  $-0.026$  to  $0.037$ ;  $p = 0.75$ ) (Supplemental Figure 2).

#### 3.2. Secondary efficacy outcomes

The remission rates between ketamine and ECT groups were not significantly different in the overall population ( $RR$ ,  $0.91$ ; 95 %  $CI$ ,  $0.47$  to  $1.74$ ;  $p = 0.769$ ;  $I^2 = 86$  %) (Supplemental Figure 3A) or among patients without psychosis ( $RR$ ,  $1.12$ ; 95 %  $CI$ ,  $0.53$  to  $2.39$ ;  $p = 0.765$ ;  $I^2 = 85$  %) (Supplemental Figure 3B). For inpatients, ECT was superior to ketamine ( $RR$ ,  $0.70$ ; 95 %  $CI$ ,  $0.55$  to  $0.90$ ;  $p = 0.004$ ;  $I^2 = 0$  %) (Supplemental Figure 3C), with a substantial reduction in heterogeneity.

In terms of change in depression scores, ECT was superior to ketamine in the overall population, with moderate effect size ( $SMD$ ,  $0.56$ ; 95 %  $CI$ ,  $0.05$  to  $1.06$ ;  $p = 0.031$ ;  $I^2 = 86$  %) (Fig. 3A) and among inpatients,

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

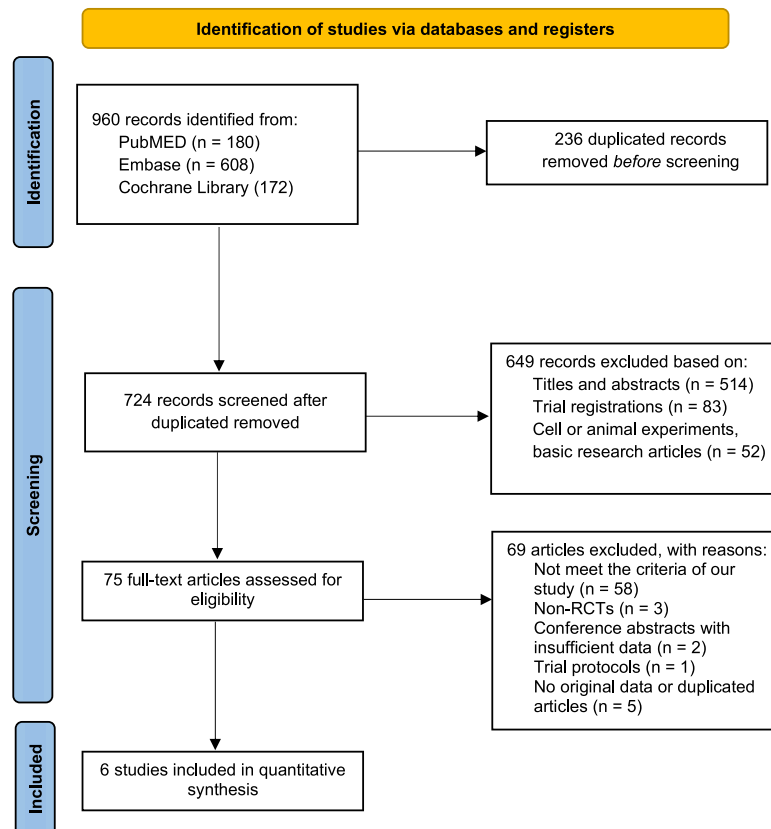


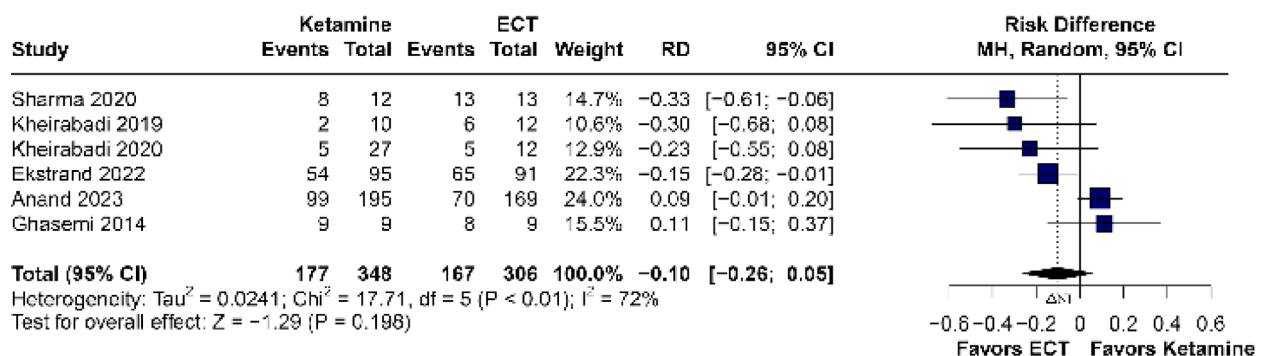
Fig. 1. PRISMA flow diagram of study screening and selection  
Abbreviations: RCT, randomized controlled trial.

**Table 1**  
Baseline patient and study characteristics.

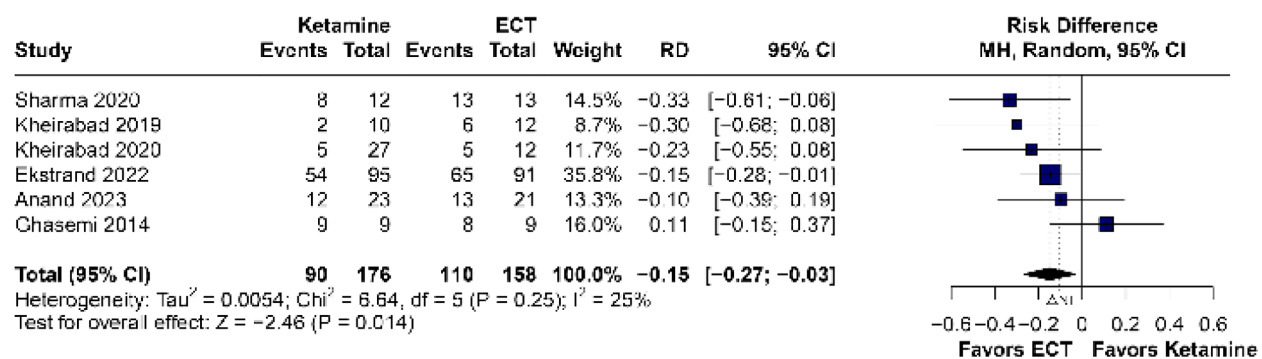
| Study and year          | Study design   | Number of patients | Male, n (%) | Age (years) | Patients without psychosis, n (%) | Admission status           | Treatment duration | Follow-up |
|-------------------------|----------------|--------------------|-------------|-------------|-----------------------------------|----------------------------|--------------------|-----------|
| Anand et al., 2023      | Open-label RCT | 365                | 174 (48)    | 46.3        | 365 (100)                         | Inpatients and outpatients | 3 weeks            | 6 months  |
| Ekstrand et al., 2022   | Open-label RCT | 186                | 67 (36)     | 52.5        | 154 (83)                          | Inpatients                 | 4 weeks            | 13 months |
| Ghasemi et al., 2014    | Blinded RCT    | 18                 | 8 (44)      | 37.6        | 18 (100)                          | Inpatients                 | 1 week             | 2 weeks   |
| Kheirabadi et al., 2019 | Blinded RCT    | 22                 | 14 (64)     | 38.8        | 22 (100)                          | Inpatients                 | 3 weeks            | 3 months  |
| Kheirabadi et al., 2020 | Open-label RCT | 39                 | 19 (49)     | 40.7        | 39 (100)                          | Inpatients                 | 3 weeks            | 7 weeks   |
| Sharma et al., 2020     | Blinded RCT    | 25                 | 11 (44)     | 38.8        | 16 (64)                           | Inpatients                 | 2 weeks            | 2 weeks   |

Abbreviation: RCT, randomized controlled trial.

### A Response rates in the overall population



### B Response rates among inpatients



**Fig. 2.** Response rates between ketamine and ECT

Caption: Forest plots presenting the risk difference (RD) and 95 % confidence interval (CI) for response rates in (A) overall population, and (B) inpatients. Abbreviations: CI, confidence interval; ECT, electroconvulsive therapy; MH, Mantel-Haenszel;  $\Delta NI$ , non-inferiority; RD, risk difference.

also with a moderate effect size ( $SMD$ , 0.71; 95 %  $CI$ , 0.38 to 1.04;  $p < 0.001$ ;  $I^2 = 30$  %) (Fig. 3B).

During follow-up, there was no significant difference in terms of relapse rates at 1 month ( $RR$ , 0.93; 95 %  $CI$ , 0.57 to 1.53;  $p = 0.781$ ;  $I^2 = 34$  %) (Supplemental Figure 4A), 3 months ( $RR$ , 0.79; 95 %  $CI$ , 0.39 to 1.61;  $p = 0.521$ ;  $I^2 = 86$  %) (Supplemental Figure 4B), or 6 months ( $RR$ , 0.99; 95 %  $CI$ , 0.69 to 1.41;  $p = 0.950$ ;  $I^2 = 40$  %) (Supplemental Figure 4C).

### 3.3. Posttreatment cognition scores

Two studies reported posttreatment cognition scores as means and dispersion measures (Anand et al., 2023; Kheirabadi et al., 2019). The scales used for quantitative synthesis were the Squire Memory

Complaint Questionnaire and the Wechsler Memory Scale, and the ketamine group was associated with better posttreatment scores than the ECT group. The effect size of this outcome was moderate ( $SMD$ , 0.50; 95 %  $CI$ , 0.30 to 0.70;  $p < 0.001$ ;  $I^2 = 0$  %) (Supplemental Figure 5).

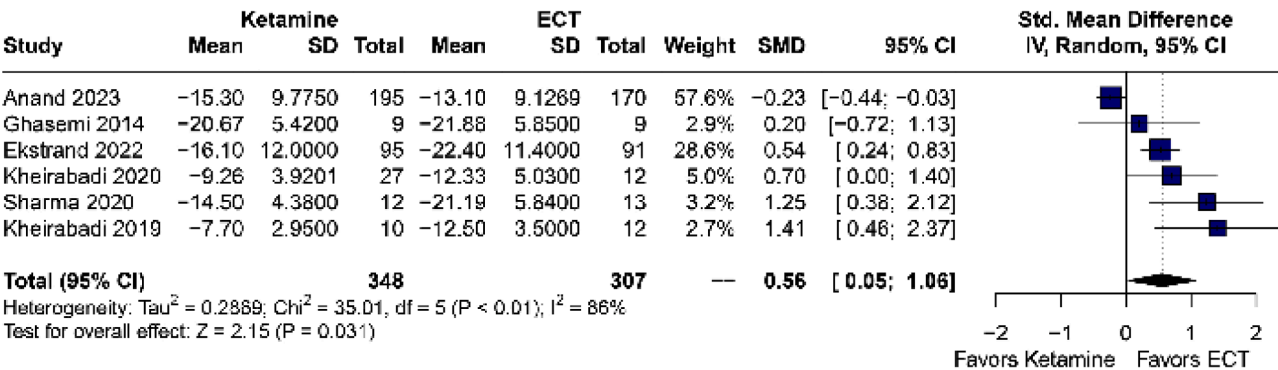
In addition, another study reported a Battery for ECT Related Cognitive Deficits (B4ECTReCoDe), however, due to the absence of dispersion measures, it was not included in the pooled analysis. In this RCT, ketamine was associated with a significant improvement in Digit Symbol Substitution Test scores, while none of the other subtests exhibited significant changes from baseline.

### 3.4. Adverse effects

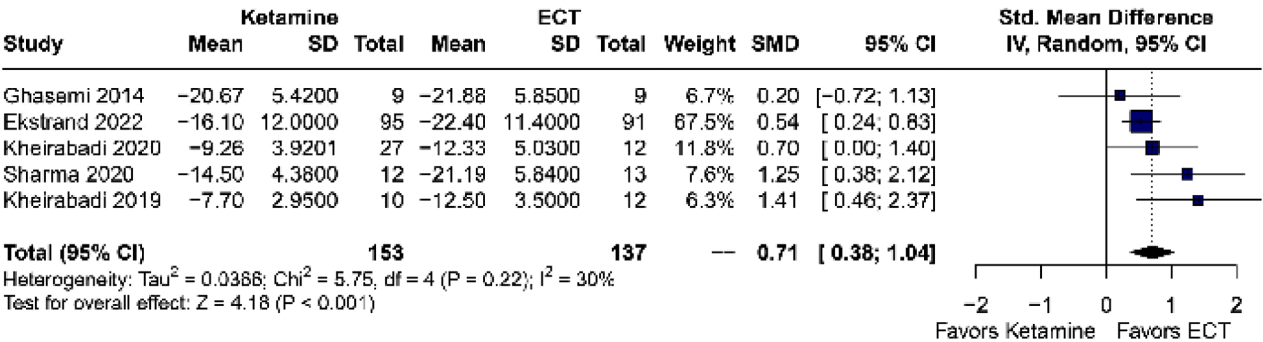
Ketamine led to a lower risk of muscle pain compared with ECT ( $RR$ ,



**A** Changes in depression scores in the overall population

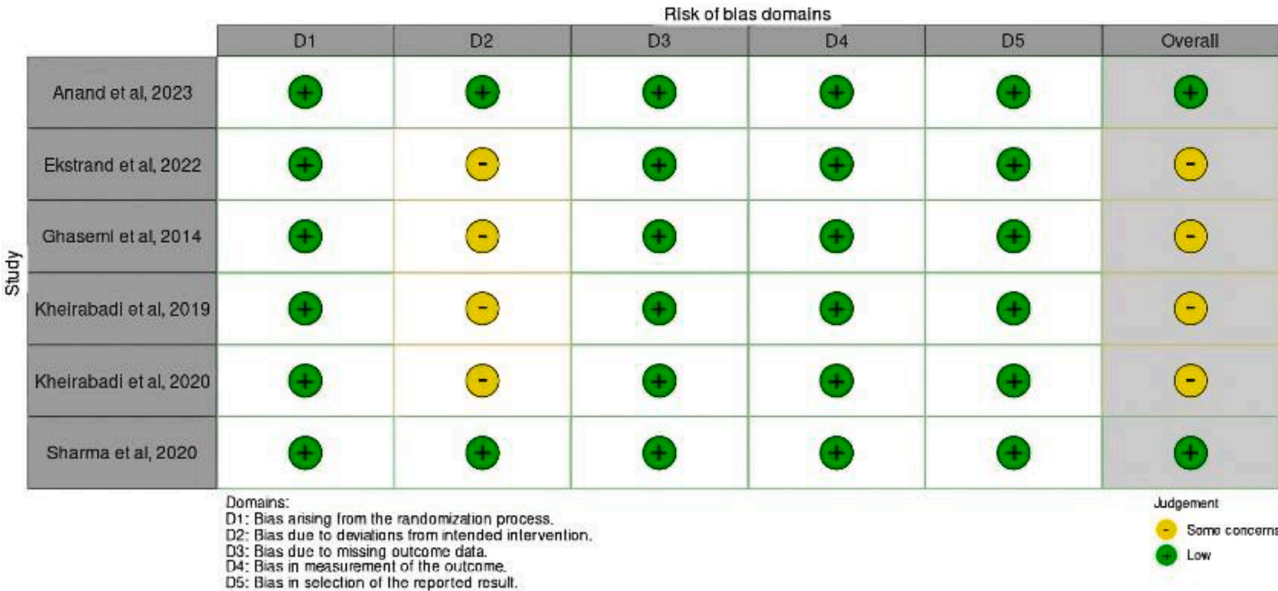


**B** Changes in depression scores among inpatients



**Fig. 3.** Changes in depression scores between ketamine and ECT  
Caption: Forest plots presenting the standardized mean difference (SMD) and 95 % confidence interval (CI) for response rates in (A) overall population, and (B) inpatients.  
Abbreviations: CI, confidence interval; ECT, electroconvulsive therapy; RD, risk difference.

0.23; 95 % CI, 0.14 to 0.39;  $p < 0.001$ ;  $I^2 = 0$  %) (Supplemental Figure 6A). However, dissociative/depersonalization symptoms were more common in patients receiving ketamine (RR, 6.03; 95 % CI, 1.94 to 18.71;  $p = 0.002$ ;  $I^2 = 14$  %) (Supplemental Figure 6B). Furthermore, there was no statistically significant difference in terms of headache incidence (RR, 0.48; 95 % CI, 0.22 to 1.03;  $p = 0.058$ ;  $I^2 = 81$  %) (Supplemental Figure 6C).



**Fig. 4.** Risk of Bias Assessment (ROB-2) of all included studies.

### 3.5. Risk of bias assessment

RoB-2 identified two RCTs at low risk of bias and four with some concerns (Fig. 4). Four RCTs had some concerns related to deviations from intended interventions (Ekstrand et al., 2022; Ghasemi et al., 2014; Kheirabadi et al., 2020, 2019). Otherwise, all domains were identified as having a low risk of bias. Moreover, the funnel plot suggested evidence of small study effects. (Supplemental Figure 7).

## 4. Discussion

In this systematic review and non-inferiority meta-analysis, we compared the efficacy and safety of ketamine and ECT for patients with MDD who were currently experiencing MDE. Our main findings were as follows: (1) the non-inferiority of ketamine to ECT was not shown in terms of response rate; (2) among inpatients, ketamine was inferior to ECT; and (3) ketamine was associated with better posttreatment cognition scores.

In various subanalyses restricted to inpatients, our findings suggested that the heterogeneity in the results may be caused by patient status at recruitment. This hypothesis is tempered by studies that showed that ECT has greater efficacy in inpatients than in outpatients (Nordenskjöld et al., 2012). However, only one RCT compared ketamine with ECT among outpatients, in which ketamine was non-inferior to ECT, supporting its use in this population (Anand et al., 2023).

Regarding safety outcomes, ketamine increased the rate of depersonalization or dissociative symptoms, but also reduced the incidence of muscle pain and improved cognitive outcomes compared with ECT. This possible protective effect of ketamine on cognition must be considered in the medical and patient treatment decision (Gill et al., 2021; Vaccarino et al., 2022).

The relapse rates of ketamine and ECT groups were similar during follow-up, with a considerable incidence. Evidence suggests that maintaining ECT reduces the risk of relapse, although this effect has not yet been established for ketamine (Berman et al., 2000; Ekstrand et al., 2022; Jelovac et al., 2013; Murrough et al., 2011; Nordenskjöld et al., 2012; Zareba et al., 2020). None of the studies in this meta-analysis compared the relative efficacy of ketamine vs. ECT continuation for relapse prevention.

The comparative efficacy of ketamine and ECT in patients with psychotic depression could not be assessed in this meta-analysis, because only one study reported stratified results for this subgroup. However, for patients without psychosis, there was no significant difference between the treatments, and non-inferiority of ketamine was not shown. This result may have yielded a greater benefit for ECT among patients with psychotic depression, considering that this characteristic predicts ECT response (Elias et al., 2018).

The influence of patient age on the comparative efficacy of ketamine and ECT remains unclear. Although a meta-regression analysis found no significant interaction between mean age and response rate, two of the included studies suggested that the ketamine efficacy compared to ECT tends to be more pronounced among younger patients (Anand et al., 2023; Ekstrand et al., 2022). This hypothesis aligns with the results of previous studies indicating an increased response to ECT among older adults (Dong et al., 2018; Elias et al., 2018). Due to the number of studies included in our meta-analysis (< 10), we cannot exclude the possibility that our analysis is underpowered to identify a significant treatment interaction with age.

Therefore, our findings support the use of ECT as first-line therapy, especially for hospitalized patients. However, for nonhospitalized patients without psychosis, ketamine may be a promising intervention, in accordance with patient preferences and tolerance to the respective adverse effects associated with these treatments. This highlights the need for further research investigating ketamine compared with ECT in community and outpatient settings, especially among young patients without psychotic features. Additionally, further studies are also needed

to evaluate ketamine and ECT continuation for the prevention of relapse.

Our study builds upon meta-analyses by incorporating data from the ELEKT-D trial (Anand et al., 2023), which is the largest RCT conducted to date and has yielded controversial results (Menon et al., 2023; Rhee et al., 2022). This data availability allowed a more robust analysis. First, we meticulously explored the heterogeneity by performing subanalyses based on psychosis and admission status, justifying the heterogeneity in our results (Shafiee et al., 2023). Second, we were able to evaluate additional outcomes, including relapse rates at 1, 3, and 6 months. Finally, performing a non-inferiority analysis allowed a more adequate assessment for this comparison, as ketamine has better availability and a safer cognitive profile than ECT (Basso et al., 2020; Ekstrand et al., 2022; Kheirabadi et al., 2020; Sackeim, 2017; Semkowska and McLoughlin, 2010; Veraart et al., 2021; Wilkinson et al., 2018).

There are several limitations in this meta-analysis. First, there were slight differences in the inclusion and exclusion criteria among studies, which we addressed by subanalyses. For instance, the trials varied in the exclusion of patients with psychotic features, and only one trial comprised outpatients. Second, there were differences in treatment protocols, which may have influenced the results, as detailed in Supplemental Table 3. Particularly, it is important to emphasize that the ELEKT-D trial used ultra-brief pulse stimulation for ECT and limited the number of sessions to 9, which may explain its considerably lower remission rates compared with other studies of this meta-analysis (Anand et al., 2023; Ekstrand et al., 2024). Third, the majority of the eligible studies had some concerns of bias and lacked long-term follow-up. Lastly, studies have assessed some outcomes at different scales, however, to mitigate this limitation, we computed SMD.

Nonetheless, this meta-analysis indicates that among patients with an MDE, ketamine did not meet the prespecified criteria for non-inferiority compared with ECT, and may be inferior among inpatients. However, recent evidence suggests ketamine as a viable alternative to ECT for outpatients. Therefore, further RCTs are needed to clarify the relative efficacy of ketamine in this patient population.

## Disclosures

The authors report no relationships that could be construed as conflicts of interest. All the authors take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation. There were no external funding sources for this study.

## CRediT authorship contribution statement

**Arthur Bezerra Cavalcanti Petrucci:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **João Vitor Andrade Fernandes:** Writing – review & editing, Investigation. **Isabelle Albuquerque Reis:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Giovanna Hanike Santos da Silva:** Supervision, Investigation. **Brenda Maria Folli Recla:** Writing – review & editing, Investigation, Data curation. **Juliana Celga de Mendonça:** Writing – review & editing, Investigation. **Victória Carvalho Souto Pedro:** Investigation. **Luís Eduardo Negreiros D'Assunção:** Writing – review & editing, Investigation. **Leandro da Costa Lane Valiengo:** Writing – review & editing, Supervision, Project administration.

## Declaration of competing interest

The authors report no relationships that could be construed as conflicts of interest. All the authors take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation. There were no external funding sources for this study.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.psychres.2024.115994](https://doi.org/10.1016/j.psychres.2024.115994).

## References

- Allen, A.P., Naughton, M., Dowling, J., Walsh, A., Ismail, F., Shorten, G., Scott, L., McLoughlin, D.M., Cryan, J.F., Dinan, T.G., Clarke, G., 2015. Serum BDNF as a peripheral biomarker of treatment-resistant depression and the rapid antidepressant response: a comparison of ketamine and ECT. *J. Affect. Disord.* 186, 306–311. <https://doi.org/10.1016/j.jad.2015.06.033>.
- American Psychiatric Association, 2013. *Diagnostic and Statistical Manual of Mental disorders*. DSM-5, 5th edition. American Psychiatric Association, Arlington, VA.
- American Psychiatric Association, 2001. *The Practice of Electroconvulsive Therapy: Recommendations for Treatment, Training, and Privileging*, 2nd Edition. American Psychiatric Publishing.
- Anand, A., Mathew, S.J., Sanacora, G., Murrough, J.W., Goes, F.S., Altinay, M., Aloysi, A. S., Asghar-Ali, A.A., Barnett, B.S., Chang, L.C., Collins, K.A., Costi, S., Iqbal, S., Jha, M.K., Krishnan, K., Malone, D.A., Nikayin, S., Nissen, S.E., Ostroff, R.B., Reti, I. M., Wilkinson, S.T., Wolski, K., Hu, B., 2023. Ketamine versus ECT for nonpsychotic treatment-resistant major depression. *N. Engl. J. Med.* 388, 2315–2325. <https://doi.org/10.1056/NEJMoa2302399>.
- Basso, L., Bönke, L., Aust, S., Gärtner, M., Heuser-Collier, I., Otte, C., Wingenfeld, K., Bajbouj, M., Grimm, S., 2020. Antidepressant and neurocognitive effects of serial ketamine administration versus ECT in depressed patients. *J. Psychiatr. Res.* 123, 1–8. <https://doi.org/10.1016/j.jpsychres.2020.01.002>.
- Berman, R.M., Cappiello, A., Anand, A., Oren, D.A., Heninger, G.R., Charney, D.S., Krystal, J.H., 2000. Antidepressant effects of ketamine in depressed patients. *Biol. Psychiatry* 47, 351–354. [https://doi.org/10.1016/S0006-3223\(99\)00230-9](https://doi.org/10.1016/S0006-3223(99)00230-9).
- Charlson, F., Siskind, D., Doi, S.A.R., McCallum, E., Broome, A., Lie, D.C., 2012. ECT efficacy and treatment course: a systematic review and meta-analysis of twice vs thrice weekly schedules. *J. Affect. Disord.* 138, 1–8. <https://doi.org/10.1016/j.jad.2011.03.039>.
- Cohen, J., 2013. *Statistical power analysis for the behavioral sciences*. Routledge.
- Dong, M., Zhu, X., Zheng, W., Li, X., Ng, C.H., Ungvari, G.S., Xiang, Y., 2018. Electroconvulsive therapy for older adult patients with major depressive disorder: a systematic review of randomized controlled trials. *Psychogeriatrics* 18, 468–475. <https://doi.org/10.1111/psyg.12359>.
- Ekstrand, J., Fattah, C., Persson, M., Cheng, T., Nordanskog, P., Åkeson, J., Tingström, A., Lindström, M.B., Nordenskjöld, A., Movahed Rad, P., 2022. Racemic ketamine as an alternative to electroconvulsive therapy for unipolar depression: a randomized, open-label, non-inferiority trial (KetECT). *Int. J. Neuropsychopharmacol.* 25, 339–349. <https://doi.org/10.1093/ijnp/pyab088>.
- Ekstrand, J., Takamiya, A., Nordenskjöld, A., Kirov, G., Sienaeert, P., Kellner, C.H., Movahed Rad, P., 2024. Ketamine or ECT? What we learned from the KetECT and ELEKT-D trials? *Int. J. Neuropsychopharmacol.* 27. <https://doi.org/10.1093/ijnp/pyad065>.
- Elias, A., Phutane, V.H., Clarke, S., Prudic, J., 2018. Electroconvulsive therapy in the continuation and maintenance treatment of depression: systematic review and meta-analyses. *Aust. N. Z. J. Psychiatry* n 415–424. <https://doi.org/10.1177/0004867417743343>.
- Espinoza, R.T., Kellner, C.H., 2022. Electroconvulsive therapy. *N. Engl. J. Med.* 386, 667–672. <https://doi.org/10.1056/NEJMr2034954>.
- Friedrich, M.J., 2017. Depression is the leading cause of disability around the world. *JAMA* 317, 1517. <https://doi.org/10.1001/jama.2017.3826>.
- Gao, M., Rejaei, D., Liu, H., 2016. Ketamine use in current clinical practice. *Acta Pharmacol. Sin.* 37, 865–872. <https://doi.org/10.1038/aps.2016.5>.
- Gelenberg, A., Freeman, M., Markowitz, J., 2010. Practice guideline for the treatment of patients with major depressive disorder. *Am. J. Psychiatry* 167, 1–118.
- Ghasemi, M., Kazemi, M.H., Yoosefi, A., Ghasemi, A., Paragomi, P., Amini, H., Afzali, M. H., 2014. Rapid antidepressant effects of repeated doses of ketamine compared with electroconvulsive therapy in hospitalized patients with major depressive disorder. *Psychiatry Res.* 215, 355–361. <https://doi.org/10.1016/j.psychres.2013.12.008>.
- Gill, H., Gill, B., Rodrigues, N.B., Lipsitz, O., Rosenblatt, J.D., El-Halabi, S., Nasri, F., Mansur, R.B., Lee, Y., McIntyre, R.S., 2021. The effects of ketamine on cognition in treatment-resistant depression: a systematic review and priority avenues for future research. *Neurosci. Biobehav. Rev.* 120, 78–85. <https://doi.org/10.1016/j.neubiorev.2020.11.020>.
- Greenberg, P.E., Fournier, A.A., Sisitsky, T., Simes, M., Berman, R., Koenigsberg, S.H., Kessler, R.C., 2021. The economic burden of adults with major depressive disorder in the United States (2010 and 2018). *Pharmacoeconomics* 39, 653–665. <https://doi.org/10.1007/s40273-021-01019-4>.
- Guo, Y., Xia, M., Ye, R., Bai, T., Wu, Y., Ji, Y., Yu, Y., Ji, G.-J., Wang, K., He, Y., Tian, Yanghua, 2024. Electroconvulsive therapy regulates brain connectome dynamics in patients with major depressive disorder.
- Gutiérrez-Rojas, L., Porras-Segovia, A., Dunne, H., Andrade-González, N., Cervilla, J.A., 2020. Prevalence and correlates of major depressive disorder: a systematic review. *Rev. Bras. Psiquiatr.* Sao Paulo Braz. 42, 657–672. <https://doi.org/10.1590/1516-4446-2020-0650>, 1999.
- Higgins, J., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M.J., Welch, V.A., 2019. *Cochrane Handbook for Systematic Reviews of Interventions*, 1st ed.
- Husain, M.M., Rush, A.J., Fink, M., Knapp, R., Petrides, G., Rummans, T., Biggs, M.M., O'Connor, K., Rasmussen, K., Little, M., Zhao, W., Bernstein, H.J., Smith, G., Mueller, M., McClintock, S.M., Bailine, S.H., Kellner, C.H., 2004. Speed of response and remission in major depressive disorder with acute electroconvulsive therapy (ECT): a Consortium for Research in ECT (CORE) report. *J. Clin. Psychiatry* 65, 485–491. <https://doi.org/10.4088/jcp.v65n0406>.
- Jelovac, A., Kolshus, E., McLoughlin, D.M., 2013. Relapse following successful electroconvulsive therapy for major depression: a meta-analysis. *Neuropsychopharmacol. Off. Publ. Am. Coll. Neuropsychopharmacol.* 38, 2467–2474. <https://doi.org/10.1038/npp.2013.149>.
- Kellner, C.H., Fink, M., Knapp, R., Petrides, G., Husain, M., Rummans, T., Mueller, M., Bernstein, H., Rasmussen, K., O'Connor, K., Smith, G., Rush, A.J., Biggs, M., McClintock, S., Bailine, S., Malur, C., 2005. Relief of expressed suicidal intent by ECT: a Consortium for Research in ECT Study. *Am. J. Psychiatry* 162, 977–982. <https://doi.org/10.1176/appi.ajp.162.5.977>.
- Kheirabadi, D., Kheirabadi, G.R., Mirlouhi, Z., Tarrahi, M.J., Norbakhsh, A., 2020. Comparison of rapid antidepressant and antisuicidal effects of intramuscular ketamine, oral ketamine, and electroconvulsive therapy in patients with major depressive disorder: a pilot study. *J. Clin. Psychopharmacol.* 40, 588–593. <https://doi.org/10.1097/JCP.0000000000001289>.
- Kheirabadi, G., Vafaie, M., Kheirabadi, D., Mirlouhi, Z., Hajiannasab, R., 2019. Comparative effect of intravenous ketamine and electroconvulsive therapy in major depression: a randomized controlled trial. *Adv. Biomed. Res.* 8, 25. <https://doi.org/10.4103/abr.abr.166.18>.
- Liberati, A., Altman, D.G., Tetzlaff, J., Mulrow, C., Gotzsche, P.C., Ioannidis, J.P.A., Clarke, M., Devereaux, P.J., Kleijnen, J., Moher, D., 2009. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ* 339, b2700. <https://doi.org/10.1136/bmj.b2700>.
- Lisanby, S.H., 2007. Electroconvulsive therapy for depression. *N. Engl. J. Med.* 357, 1939–1945. <https://doi.org/10.1056/NEJMc075234>.
- Loureiro, J.R.A., Leaver, A., Vasavada, M., Sahib, A.K., Kubicki, A., Joshi, S., Woods, R. P., Wade, B., Congdon, E., Espinoza, R., Narr, K.L., 2020. Modulation of amygdala reactivity following rapidly acting interventions for major depression. *Hum. Brain Mapp.* 41, 1699–1710. <https://doi.org/10.1002/hbm.24895>.
- McKibbin, K.A., Wilczynski, N.L., Haynes, R.B., 2009. Retrieving randomized controlled trials from MEDLINE: a comparison of 38 published search filters. *Health Inf. Libr. J.* 26, 187–202. <https://doi.org/10.1111/j.1471-1842.2008.00827.x>.
- Menon, V., Varadharajan, N., Faheem, A., Andrade, C., 2023. Ketamine vs electroconvulsive therapy for major depressive episode: a systematic review and meta-analysis. *JAMA Psychiatry* 80, 639–642. <https://doi.org/10.1001/jamapsychiatry.2023.0562>.
- Murrough, J.W., Perez, A.M., Mathew, S.J., Charney, D.S., 2011. A case of sustained remission following an acute course of ketamine in treatment-resistant depression. *J. Clin. Psychiatry* 72, 2408. <https://doi.org/10.4088/JCP.10106447blu>.
- Nikayin, S., Rhee, T.G., Cunningham, M.E., de Fontnouvelle, C.A., Ostroff, R.B., Sanacora, G., Wilkinson, S.T., 2022. Evaluation of the trajectory of depression severity with ketamine and esketamine treatment in a clinical setting. *JAMA Psychiatry* 79, 736–738. <https://doi.org/10.1001/jamapsychiatry.2022.1074>.
- Nordenskjöld, A., von Knorring, L., Engström, I., 2012. Predictors of the short-term responder rate of Electroconvulsive therapy in depressive disorders—a population based study. *BMC Psychiatry* 12, 115. <https://doi.org/10.1186/1471-244X-12-115>.
- Rasmussen, K.G., Lineberry, T.W., Galardy, C.W., Kung, S., Lapid, M.I., Palmer, B.A., Ritter, M.J., Schak, K.M., Sola, C.L., Hanson, A.J., Frye, M.A., 2013. Serial infusions of low-dose ketamine for major depression. *J. Psychopharmacol. Oxf. Engl.* 27, 444–450. <https://doi.org/10.1177/0269881113478283>.
- Rhee, T.G., Shim, S.R., Forester, B.P., Nierenberg, A.A., McIntyre, R.S., Papakostas, G.I., Krystal, J.H., Sanacora, G., Wilkinson, S.T., 2022. Efficacy and safety of ketamine vs electroconvulsive therapy among patients with major depressive episode: a systematic review and meta-analysis. *JAMA Psychiatry* 79, 1162–1172. <https://doi.org/10.1001/jamapsychiatry.2022.3352>.
- Rush, A.J., Trivedi, M.H., Wisniewski, S.R., Nierenberg, A.A., Stewart, J.W., Warden, D., Niederrehe, G., Thase, M.E., Lavori, P.W., Lebowitz, B.D., McGrath, P.J., Rosenbaum, J.F., Sackeim, H.A., 2006. Acute and longer-term outcomes in depressed outpatients requiring one or several treatment steps: a STAR\*D report. *Am. J. Psychiatry*.
- Sackeim, H.A., 2017. Modern electroconvulsive therapy: vastly improved yet greatly underused. *JAMA Psychiatry* 74, 779–780. <https://doi.org/10.1001/jamapsychiatry.2017.1670>.
- Semkovska, M., McLoughlin, D.M., 2010. Objective cognitive performance associated with electroconvulsive therapy for depression: a systematic review and meta-analysis. *Biol. Psychiatry* 68, 568–577. <https://doi.org/10.1016/j.biopsych.2010.06.009>.
- Shafiee, A., Soltani Abhari, F., Jafarabady, K., Bakhtiari, M., 2023. Ketamine versus electroconvulsive therapy for treatment-resistant depression: an updated meta-analysis of randomized clinical trials. *Asian J. Psychiatry* 88, 103720. <https://doi.org/10.1016/j.ajp.2023.103720>.
- Sharma, R.K., Kulkarni, G., Kumar, C.N., Arumugham, S.S., Sudhir, V., Mehta, U.M., Mitra, S., Thanki, M.V., Thirithalli, J., 2020. Antidepressant effects of ketamine and ECT: a pilot comparison. *J. Affect. Disord.* 276, 260–266. <https://doi.org/10.1016/j.jad.2020.07.066>.
- Shim, S.R., Kim, S.-J., 2019. Intervention meta-analysis: application and practice using R software. *Epidemiol. Health* 41, e2019008. <https://doi.org/10.4178/epih.e2019008>.
- Smith-Apeldoorn, S.Y., Veraart, J.K., Spijker, J., Kamphuis, J., Schoevers, R.A., 2022. Maintenance ketamine treatment for depression: a systematic review of efficacy, safety, and tolerability. *Lancet Psychiatry* 9, 907–921. [https://doi.org/10.1016/S2215-0366\(22\)00317-0](https://doi.org/10.1016/S2215-0366(22)00317-0).

- Sterne, J.A.C., Savović, J., Page, M.J., Elbers, R.G., Blencowe, N.S., Boutron, I., Cates, C. J., Cheng, H.-Y., Corbett, M.S., Eldridge, S.M., Emberson, J.R., Hernán, M.A., Hopewell, S., Hróbjartsson, A., Junqueira, D.R., Jüni, P., Kirkham, J.J., Lasserson, T., Li, T., McAleenan, A., Reeves, B.C., Shepperd, S., Shrier, I., Stewart, L.A., Tilling, K., White, I.R., Whiting, P.F., Higgins, J.P.T., 2019. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. <https://doi.org/10.1136/bmj.14898> 14898.
- Sterne, J.A.C., Sutton, A.J., Ioannidis, J.P.A., Terrin, N., Jones, D.R., Lau, J., Carpenter, J., Rucker, G., Harbord, R.M., Schmid, C.H., Tetzlaff, J., Deeks, J.J., Peters, J., Macaskill, P., Schwarzer, G., Duval, S., Altman, D.G., Moher, D., Higgins, J.P.T., 2011. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* 343, d4002. <https://doi.org/10.1136/bmj.d4002>. –d4002.
- The Handbook of Research Synthesis and Meta-Analysis, 2019. Russell Sage Foundation. <https://doi.org/10.7758/9781610448864>.
- Vaccarino, S., Adamsahib, F., Milev, R., Parikh, S., Lam, R., Blier, P., Kennedy, S., Ladha, K., Bhat, V., 2022. The effects of ketamine on cognition in unipolar and bipolar depression: a systematic review. *Eff. Ketamine Cogn. Unipolar Bipolar Depress. Syst. Rev.*
- Veraart, J.K.E., Smith-Apeldoorn, S.Y., Spaans, H.-P., Kamphuis, J., Schoevers, R.A., 2021. Is ketamine an appropriate alternative to ECT for patients with treatment resistant depression? A systematic review. *J. Affect. Disord.* 281, 82–89. <https://doi.org/10.1016/j.jad.2020.11.123>.
- Wilkinson, S.T., Agbese, E., Leslie, D.L., Rosenheck, R.A., 2018. Identifying recipients of electroconvulsive therapy: data from privately insured americans. *Psychiatr. Serv.* 69, 542–548. <https://doi.org/10.1176/appi.ps.201700364>.
- Zareba, K., Banasiewicz, J., Rozenek, H., Wójtowicz, S., Jakiel, G., 2020. Peripartum predictors of the risk of postpartum depressive disorder: results of a case-control study. *Int J. Env. Res Public Health Online* 17.